

Chromatic number of $\frac{3}{3}$ -power of graph

OPG-60055

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let G be a graph and $m, n \in \mathbb{N}$. The graph G_n^m is defined to be the m -power of the n -subdivision of G . In other words, $G_n^m = (G_n^1)^m$.

Conjecture Let G be a graph with $\Delta(G) \geq 2$. Then $\chi(G^{\frac{3}{3}}) \leq 2\Delta(G) + 1$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>